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Supplementary Material: Agentic Autoresearch for CT Reconstruction

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ABSTRACT

This document contains the supplementary material for *Agentic Autoresearch for CT Reconstruction*. It provides the full solver taxonomy and per-method configuration (S1), the Mayo-LDCT data, split, and geometry pipeline (S2), the breast split and train/test leakage audit (S3), the Mayo test board and significance (S4), the parameter-efficient recombination study (S5), the framework-axis reading of the breast reversal and the noise protocol (S6; the full boards are in the main text), an extended account of agent behavior (S7), and the list of additional supplementary figures (S8).

TABLE S1 | Full solver taxonomy. Axis A = physics engagement; Axis B = prior source. The final column cites the source(s) for each family. Abbreviations: DC, data consistency; N2I, Noise2Inverse; rectified-flow prior, a fast diffusion-type generative prior; FoE, field of experts (a learned filter-bank image prior).

Family	Representative solver(s)	Data-consistency D	Prior $\mathcal{R}$	Axis A	Axis B	Ref.
Classical iterative	tv-iterative, manhart-PWLS-TV	filtered / SART grad	total variation	in-loop	hand-crafted	[1,2]
Image-domain denoiser	manduca-bilateral, DD-UNet sup.	none (post-FBP)	bilateral / U-Net	none	hand-cr. / sup.	[3,4]
Unrolled learned-iterative	ITNet v1/v2/v3, U-Swin	in-loop DC	learned CNN / transformer	in-loop	supervised	[5,6,7,8,9]
Variational network	Hammernik-2017, Hammernik-VN	in-loop DC	field-of-experts	in-loop	supervised	[10,11]
Learned primal-dual	learned-primal-dual (LPD)	filtered + learned dual	learned combination	in-loop	supervised	[12]
Dual-domain	DD-supervised, DD-bilateral, DD-N2I	image + sinogram	sup. / bilateral / self-sup.	single/in-loop	sup. / self-sup.	[13,14]
Generative prior	fastdiff flow/wavelet ×4	DC + diffusion	rectified-flow prior	in-loop	generative	[15,16,17,18]
Per-scene implicit	NAF, R^2-Gaussian	per-scene fit	implicit representation	per-scene	implicit	[19,20,21]
Foundation zero-shot	RAM	single DC	frozen foundation model	single	foundation	[22]
Band decomposition	Wu-2015, Wu-2015-trainable	band-weighted DC	frequency-band residual	in-loop	hand-cr. / sup.	[2]
Recombination (ours)	**param-efficient**	**filtered + LPD combine**	**bilateral post-filter**	**in-loop**	**hybrid**	—

S1 | Solver taxonomy and per-method configuration (Table 1, full)

We place the 26 methods on two axes. Axis A is physics engagement (how the operator **A** is used at inference). Axis B is the prior source $\mathcal{R}$.

Full per-solver `CONFIG` defaults, parameter counts, and design notes are in each solver's design doc under `pentathlon/demo_dl_reference/` and the observation records under `docs/runs/`.

S2 | Mayo-LDCT — data, split, and the geometry pipeline

Real helical low-dose CT from the 2016 AAPM Low-Dose CT Grand Challenge.[23] We rebinned helical to fan-beam by single-slice rebinning.[24] The differentiable fan-beam projector used at every step is PYRO-NN.[25] This pipeline was the dominant human cost of the whole study (~24 active working days; three travel gaps excluded). We use a fixed patient split: train L145/186/209/219, validation L277, test L014/056/058/075/123. Test scores are the per-patient mean ± std over the five test patients ($n = 5$). Geometry details, rebinning validation, and the effort timeline figure are in the repository (`docs/runs/mayo_effort_timeline.png`). The Mayo challenge is one of a family of AAPM reconstruction grand challenges spanning sparse-view, spectral, truth-based, and metal-artifact tasks.[26,27,28,29,30]

S3 | Breast split and the train/test leakage audit

Split. We re-partitioned the public 4000-case Sidky train pool into train (3600), validation (200), and test (200). The split is deterministic (seed 20260703) and disjoint. The official challenge val/test truths are withheld, so we cannot use them; our test split is drawn from the public pool.

TABLE S2 | Breast versus Mayo: the two problems are limited by different things, so their absolute hr scales are not comparable.

	Breast (Sidky)	**Mayo**
data	noiseless, ideal	real low-dose, quant
bottleneck	data incompleteness	noise / dose
RMSE floor	0	> 0
best hr	~0.89	~0.38
dominant lever	data-consistency / known operator	denoising

TABLE S3 | Parameter-efficient recombination arc on breast (128-view sparse), all under the 20-min budget.

step	**architecture**	**params**	**val hr**
image-domain unroll (any config)	Mayo-style FoE denoiser	~1930	≤ 0.26 (ceiling)
filtered DC FBP(**Ax** − **g**)	$K = 30$	**13**	**0.4345**
+ learned primal-dual combination	$W = 4$	183	0.5047
+ **3-scale bilateral output filter**	$\sigma_r = 0.02$	**195**	**0.6201** (single best)

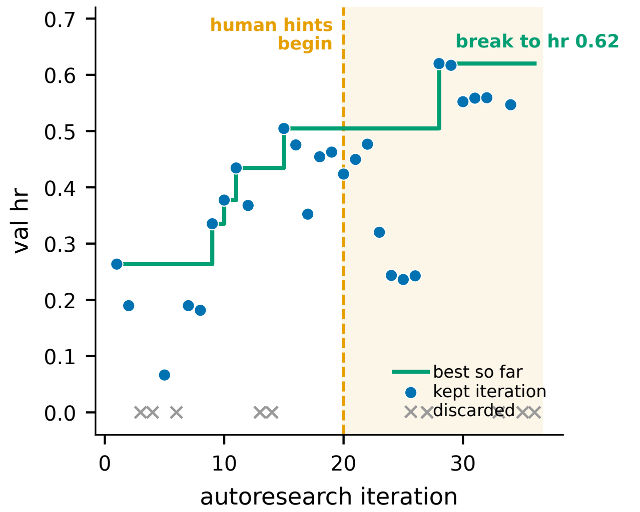

FIGURE S1 | Compact-solver search trajectory (breast, 128-view sparse). Points are per-iteration validation headroom (discarded iterations shown at hr 0); the step line is the best value so far. The agent's own first 20 iterations plateau at hr ≈ 0.50; the decisive break to hr 0.62 comes only in the hinted second half (shaded), at iteration 28.

Leakage audit (three independent checks).

1. *Mechanism.* Training loads the `train` split. The test redirect fires only for the evaluation load. Training never sees test.
2. *Pixel-hash disjointness.* Over all 3600 train, 200 val, and 200 test truth images: TEST∩TRAIN = 0, TEST∩VAL = 0, VAL∩TRAIN = 0.
3. *Behavioral.* Test-hr tracks val-hr for every method, within ±0.01, and several methods score *lower* on test than val. Training on test would make test-hr much higher. It does not. The high absolute hr is the dataset (noiseless, well-posed), not a leak.

Why breast hr is high, and not comparable to Mayo. The Sidky data is noiseless by design; its RMSE floor is zero. Breast is incompleteness-limited (128 of ~1000 views). Mayo is noise-limited (real quantum noise). The two absolute hr scales are therefore not comparable.

S4 | Mayo test board and significance

Champion ITNet v1, test hr 0.376 ($n = 5$). The top tier is a statistical tie: ITNet v1 / ITNet v2 / U-Swin at the 5 % level; the compact param-efficient solver joins at the 1 % level ($p = 0.02$–0.027). At $n = 5$ the paired t-test has low power (a Wilcoxon test cannot even reach $p < 0.05$). The full board (all 26 solvers, hr/SSIM/RMSE) is in the main text; per-patient hr and the pairwise significance matrix are in `docs/runs/mayo_significance_stats.md` and the forest/matrix figures.

S5 | The parameter-efficient recombination study (search arc)

The agent recombined the method parts into a minimal-parameter solver. The climb was made of genuine six-box iterations. Key steps on breast (128-view sparse), all under the 20-min budget:

Findings, each earned by an iteration: the image-domain denoiser caps at ~0.26; deeper *adjoint* data-consistency lowers hr (it re-injects streaks); the *filtered* gradient breaks the ceiling at 13 parameters; an in-loop prox (proximal / denoising step) hurts (the filtered gradient already reconstructs); a learned cross-step combination helps; a full conjugate-gradient DC solve does not beat the single filtered gradient under a fixed wall; and Wu-2015 band machinery adds nothing here. The champion is seed-fragile in the variational-network manner: 4 of 5 seeds cluster at 0.571 ± 0.033, one collapses; two stabilizers did not recover it. On the held-out test set the compact solver scores hr 0.6212±0.0076 — the tightest per-case std of any solver.

The gains are back-loaded, and human-cued. They are not spread evenly across the 40-iteration search. Under the agent alone the best value plateaued at hr ≈ 0.50 by iteration 15 and did not move for a dozen iterations. Only in the second half — after a human began suggesting untried directions — did it break through to hr 0.62, at iteration 28 (Figure S1). This is the clearest single case in the study of the agent needing human steering to escape a local optimum: without the hints it would have reported the 195-parameter design at its ~0.50 plateau.

Mayo vs breast contrast. On Mayo the compact optimum is an evolved field-of-experts plus multi-scale bilateral denoiser (hr 0.324 at ~0.00097 M params), statistically tied with the top tier at the 1 % level. On a denoising problem, a tiny denoiser suffices. On sparse-view breast, that same denoiser fails; data-consistency is the differentiator.[12] This problem-dependence is the practical face of the known-operator principle[31,32]: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it curbs the hallucination risk of unconstrained learned maps.[33,34] The agent re-derived each optimum; it did not transfer.

Human role in this study. The recombination *strategy* was a human idea, on both datasets. On Mayo the human repeatedly asked whether more parameter-efficient approaches existed and explicitly suggested combining the pieces. On breast the finale was framed as a cross-method recombination and steered away from copying Mayo. The agent executed and mechanistically refined; it did not originate the recombination strategy.

S6 | Breast reversal — axes and noise protocol

The full noiseless and noisy breast boards (all 25 solvers each) are in the main text, in the table that shows the two boards side by side. This section records how the reversal reads off the framework axes and the exact noise protocol behind it.

Reading the reversal by framework axes. Robust: in-loop physics (learned-primal-dual) or hand-crafted smoothing priors (bilateral, total variation). Brittle: supervised image-domain maps trained on clean FBP (dual-domain-supervised; the constrained fastdiff variants). The hr metric is fair *within* the noisy board — all methods share the same noisy FBP baseline. The absolute SSIM/PSNR columns confirm the ranking is genuine, not baseline-gaming.

S6.1 | Noise model and no-retrain protocol

Poisson photon noise per Eq. (6) of the main text, $I_0 = 10^5$, applied to the 200 test sinograms with a fixed seed (deterministic; every method sees the identical noisy input). RMS of the sinogram perturbation $\approx$ 0.012 (line-integral units), $\sim$1.8 % at the thickest ray. Supervised methods load their noiseless checkpoint and skip training; per-scene/classical methods re-fit on the noisy input. The FBP baseline is recomputed from the noisy sinogram.

S7 | Agent-behavior — extended account

This account follows the autoresearch paradigm of automated open-ended scientific discovery[35] realized through LLM code agents that resolve real software tasks[36,37] and physics-informed agentic development in adjacent imaging fields.[38]

Strengths. Fast paper→code; strong hyper-parameter optimization; strict fixed-budget discipline; scales evaluation across many parallel benchmarks; follows well-decomposed instructions.

Weaknesses. Does not invent new methods. Must be forced to recombine (converges early and pads if unsupervised — one breast sub-run replayed identical configs until an audit caught it). CT-image vision misses obvious artifacts, so numbers are the source of truth. Long tasks must be decomposed or even a 1M-token context is exhausted. Overfits to the task/metric (not only an agent trait).

Human meta-strategies (repeatable, same on both datasets). Persistence/breadth (keep asking for more directions); recombination (the idea to assemble pieces); redirection (do not mimic the other dataset); auditing (catch padding, enforce budget, fix provenance). The defensible claim: agent = tireless, mechanistically reasoning, self-modifying executor; human = strategist + auditor.

S8 | Additional figures (supplement)

- S-Fig 1: per-iteration val-vs-test hr trajectory (selection honesty).
- S-Fig 2: Mayo significance forest plot + solver×metric matrix.
- S-Fig 3: breast noiseless vs noisy example panels for all top-10 solvers.
- S-Fig 4: parameter-efficient climb (params-vs-hr Pareto), both datasets.
- S-Fig 5: effect-size (Cohen d_z) vs. raw Δhr, illustrating the $n = 5$ vs $n = 200$ contrast.

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